

UnitScope Supplementary Material

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1 Complete Formal Proofs

This section establishes the supplementary Snapshot Indistinguishability result and gives the complete proofs behind the theorem sketches in the main paper. Theorem S1 supplies the full construction for the Snapshot Indistinguishability theorem in the main paper. Each remaining proof identifies the corresponding main theorem by name. All results use addition-or-removal natural-person adjacency.

1.1 Snapshot indistinguishability under a fixed plan compiler

Fix $C > 0$. For either resolver below, use the same downstream compiler. Each resolver block G becomes one plan unit with member set G , radius C , a common execution mode, and the canonical label defined by the main paper’s descriptor commitment. The canonical-label schema is identical for both resolvers. Each resolver’s algorithm hash is fixed across its two neighboring inputs, as required for cancellation. The hashes of two different resolver programs need not be equal because workload-weighted unit reorganization (WUR) is evaluated within, not across, resolvers.

THEOREM S1 (SNAPSHOT INDISTINGUISHABILITY). *For every $K > 0$, there exist natural-person neighbors $X \simeq_{u_0} X'$, a deletion-closed natural-person input domain, and deterministic resolvers L_{stable} and L_{trigger} such that both resolvers output the same true partition on X , and hence both have pairwise $F1 = 1$ and identical realized-partition statistics. Use one compiler that maps every resolver block to a radius- C unit. For both resolvers, canonical labels commit to the task domain, epoch, execution mode, radius, member set, resolver algorithm hash, and coupled-randomness identifier. The stable resolver has a global WUR envelope C on that domain, while this common compiler gives*

$$\sup_v \sup_{Y \simeq_v Y'} \text{WUR}_{\text{stable}}(Y, Y') = C, \quad \text{WUR}_{\text{trigger}}(X, X') = (2n + 1)C \geq KC \quad (1)$$

for some n depending on K .

PROOF. Choose $n \geq \max\{2, \lceil (K - 1)/2 \rceil\}$. For each $i \in \{0, \dots, n\}$, natural person u_i owns exactly the two records $V_i = \{a_i, b_i\}$. Let the input domain contain $Y_S = \bigcup_{i \in S} V_i$ for every $S \subseteq \{0, \dots, n\}$. This domain is closed under natural-person deletion, and the true partition of Y_S is $\{V_i : i \in S\}$.

The stable resolver always returns this true partition. Under any natural-person deletion, every surviving block has the same member commitment, radius, execution mode, algorithm hash, and randomness identifier on both sides. Those canonical labels cancel, leaving exactly the deleted person’s radius- C label. Thus every nontrivial stable transition has WUR C , which proves the displayed global envelope.

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The trigger resolver also returns the true partition whenever u_0 is present. When u_0 is absent and the surviving indices are $i_1 < \dots < i_m$, with $m \geq 2$, it returns the cyclic re-pairing

$$W_j = \{a_{i_j}, b_{i_{j+1}}\} \quad (1 \leq j \leq m), \quad i_{m+1} = i_1. \quad (2)$$

For zero or one survivor it returns the empty or single-block true partition.

Take $X = Y_{\{0, \dots, n\}}$ and $X' = Y_{\{1, \dots, n\}}$. Both resolvers return $\{V_0, \dots, V_n\}$ on X , which is exactly the true partition. Every true block has two records, so the pairwise metric has true-positive pairs and no false-positive or false-negative pair. Its precision, recall, and $F1$ are all one.

On X' , the trigger resolver returns n cyclic blocks. Because $n \geq 2$, no new block equals an old V_i . Membership is part of the canonical commitment, so the trigger transition has no common label. All $n + 1$ old labels and all n new labels remain after multiset cancellation. Every unit has radius C , and the existing WUR definition gives $(n + 1)C + nC = (2n + 1)C \geq KC$. $\square$

The construction also supplies the separation used in the main paper. Both resolvers expose the same partition on X , while their neighboring WUR bounds differ by an arbitrarily large factor. The Least Sound Envelope theorem in the main paper transfers each fixed-pair WUR value to the saturating query class.

1.2 Proof of Plan-Composition Sensitivity

This subsection proves the plan-composition sensitivity theorem stated in the main paper.

For a compiled plan on input X , write T_X for the finite multiset of canonical labels and $z_\tau(X)$ for the released, pre-noise vector of label τ . Its public radius is $c_\tau(X)$ and $\|z_\tau(X)\|_2 \leq c_\tau(X)$. Multiset difference and intersection retain multiplicity. Coupling fixes the public model, prior releases, and data-independent randomness.

LEMMA S2 (CANONICAL-LABEL CANCELLATION). *For two finite compiled plans, let T_X and $T_{X'}$ be their canonical-label multisets and let $z_\tau(Y)$ be the pre-noise vector of label τ on input Y . If $I = T_X \cap T_{X'}$ and equal canonical labels commit to equal members, radii, algorithm hash, and randomness identifier under coupled execution, then*

$$\sum_{\tau \in T_X} z_\tau(X) - \sum_{\tau \in T_{X'}} z_\tau(X') = \sum_{\tau \in T_X \setminus I} z_\tau(X) - \sum_{\tau \in T_{X'} \setminus I} z_\tau(X'). \quad (3)$$

PROOF. For each occurrence of a label in I , the commitment rule and coupling imply $z_\tau(X) = z_\tau(X')$. Pair equal occurrences and subtract them. What remains is exactly the two multiset differences in Eq. (3). $\square$

LEMMA S3 (RADIUS SUM AFTER CANCELLATION). *For two finite compiled plans, let $T_X, T_{X'}, I$, and $z_\tau(Y)$ satisfy $I = T_X \cap T_{X'}$. Suppose equal canonical labels have equal outputs under coupled execution and every label has $\|z_\tau(Y)\|_2 \leq c_\tau(Y)$. Then*

$$\begin{aligned} & \left\| \sum_{\tau \in T_X} z_\tau(X) - \sum_{\tau \in T_{X'}} z_\tau(X') \right\|_2 \\ & \leq \sum_{\tau \in T_X \setminus I} c_\tau(X) + \sum_{\tau \in T_{X'} \setminus I} c_\tau(X'). \end{aligned} \quad (4)$$

PROOF. Apply Eq. (3), the triangle inequality, and then the per-label clipping bound to every old-only and new-only term. No assumption about the directions of the vectors is needed. $\square$

Completion of the proof. By definition, the right side of Eq. (4) is precisely $\text{WUR}_C(X, X')$. Lemma S3 therefore gives the inequality in the Plan-Composition Sensitivity theorem in the main paper. Taking a supremum over every person and every admissible neighboring pair proves that a global WUR envelope Δ implies one-round ℓ_2 sensitivity at most Δ .

1.3 Proof of the Least Sound Envelope

This subsection proves the least sound envelope theorem stated in the main paper.

Fix one neighboring pair and its two compiled plans. The class $\mathcal{Q}_{\text{sat}}$ consists of bounded-vector queries for which every changed canonical unit can independently choose any vector in its closed radius ball. Unchanged canonical units remain coupled. This is a query-class definition, not an assumption about learned gradients.

LEMMA S4 (ALIGNED SATURATION WITNESS). *Let $A = T_X \setminus (T_X \cap T_{X'})$ and $A' = T_{X'} \setminus (T_X \cap T_{X'})$ for a fixed neighboring pair of finite compiled plans. Let $\mathcal{Q}_{\text{sat}}$ contain the bounded-vector queries in which changed units independently attain any vector in their closed radius balls and unchanged units remain coupled. For any unit vector v , there is a query in $\mathcal{Q}_{\text{sat}}$ satisfying*

$$z_\tau(X) = c_\tau(X)v \quad (\tau \in A), \quad z_\tau(X') = -c_\tau(X')v \quad (\tau \in A'). \quad (5)$$

PROOF. Independent saturation is the defining closure property of $\mathcal{Q}_{\text{sat}}$. Choose the displayed boundary point in each old-only radius ball and its antipodal boundary point in each new-only ball. Assign any common coupled value to unchanged units. Those values cancel. $\square$

Completion of the proof. The Plan-Composition Sensitivity theorem in the main paper proves the upper bound for every bounded query, hence also for this subclass. For the reverse inequality, use the query from Lemma S4. After unchanged labels cancel, its difference is

$$\begin{aligned} & \sum_{\tau \in A} c_\tau(X)v - \sum_{\tau \in A'} [-c_\tau(X')v] \\ &= \left(\sum_{\tau \in A} c_\tau(X) + \sum_{\tau \in A'} c_\tau(X') \right) v. \end{aligned} \quad (6)$$

Its norm is the scalar in parentheses because $\|v\|_2 = 1$, and that scalar is WUR. Thus the supremum attains the upper bound in the Least Sound Envelope theorem in the main paper.

Relation to actual federated learning (FL) gradients. Per-example and per-unit gradients in a concrete training round are coupled through one model state, one loss, the observed features and labels, and the grouping rule. They generally cannot choose independent directions or simultaneously attain every clipping radius. Consequently The Least Sound Envelope theorem in the main paper gives equality for $\mathcal{Q}_{\text{sat}}$, whereas for the actual FL gradient query the Plan-Composition Sensitivity theorem supplies an upper bound that may be strict. Neither an observed WUR nor its ratio to C is a realized privacy-loss multiplier.

1.4 Proof of Evidence-Graded Compiler Soundness

This subsection proves the evidence-graded compiler soundness theorem stated in the main paper.

Let $\mathcal{F}_u$ be the permanent fallback labels that person u may touch, with $|\mathcal{F}_u| \leq B$. A pure label has only u 's records whenever it is affected by removal of u . A mixed label can also contain records of other people. Its canonical commitment includes its active members.

LEMMA S5 (MIXED-LABEL TWO-SIDED CHARGE). *Fix natural-person neighbors $X \simeq_u X'$ and stable fallback radii $c_{\text{atom}}, c_{\text{mixed}} \geq 0$. Removal of u changes a pure affected fallback label by WUR at most c_{atom} , and changes a mixed affected fallback label by WUR at most $2c_{\text{mixed}}$.*

PROOF. For a pure label, removing u leaves no successor contribution, so only the old label is charged. For a mixed label, other members may remain. Because membership is part of the canonical commitment, the old and successor labels are distinct, and each is clipped at c_{mixed} . WUR therefore charges both sides. This also covers an arbitrary reversal of their output directions. $\square$

LEMMA S6 (NONREDISTRIBUTING FAIL-CLOSED DEGRADATION). *Fix an accounting epoch, permanent partial slots, permanent fallback labels, and their radii before data are read. Suppose a declaration that is invalid at epoch validation is never installed. Its records use their already registered fallback labels, no empty partial or local slot is refilled, and no unused radius is reassigned. Then the invalid declaration creates no label transition beyond the governed fallback transitions already charged for the affected person.*

PROOF. The invalid claim has no canonical plan label on either neighboring side. Every accepted record is deterministically mapped to its permanent fallback label. Labels outside $\mathcal{F}_u$ have identical member commitments and cancel. Lemma S5 accounts for the remaining labels. Because capacity is not reissued, invalidation cannot activate a label owned by another person or increase any radius. $\square$

Runtime events are handled by the same rule only when their branch and permanent successor mapping were frozen into the epoch. A replay, expired epoch, quota violation, overlapping claim, or unregistered claim is rejected. Institution nonresponse or a single invalid handle may use the preallocated fallback branch without refill. A membership or slot-policy change begins a new epoch. Treating a private-data-dependent validity change as if it were a harmless in-epoch toggle would not satisfy Lemma S6.

LEMMA S7 (NONCOMPLETE GOVERNED BUDGET). *Fix natural-person neighbors $X \simeq_u X'$ and nonnegative route radii. For a noncomplete person with $h \leq H$ valid partial groups, a affected pure labels, and m affected mixed labels, where $a + m \leq B$,*

$$\text{WUR}_C(X, X') \leq hc_{\text{ph}} + ac_{\text{atom}} + 2mc_{\text{mixed}}. \quad (7)$$

PROOF. Purity and exclusivity make every affected partial group old-only under person removal, so it costs at most c_{ph} . Quota enforcement gives at most H such groups. Permanent fallback locality leaves labels outside $\mathcal{F}_u$ unchanged. Apply Lemma S5 to the $a + m$ affected fallback labels and sum the disjoint charges. The reverse addition direction is symmetric. $\square$

Completion of the proof. If a valid complete declaration applies, exclusive extraction produces one radius- C unit and no partial or fallback unit for that person. Its removal therefore costs at most C . Otherwise apply Lemma S7 and the radii in the Evidence-Graded Compiler Soundness theorem in the main paper.

$$\begin{aligned} \text{WUR} &\leq h \frac{\lambda C}{H} + a \frac{(1-\lambda)C}{B} + 2m \frac{(1-\lambda)C}{2B} \\ &\leq \lambda C + (a+m) \frac{(1-\lambda)C}{B} \leq C. \end{aligned} \quad (8)$$

The same calculation covers a frozen fail-closed fallback state because it does not redistribute capacity. The Plan-Composition Sensitivity theorem converts the WUR bound into the numerical sensitivity bound.

1.5 Why Partial-Handle purity is necessary

LEMMA S8 (EXPLICIT ERRONEOUS-MATCH COUNTEREXAMPLE). *If a purported Partial-Handle may mix records of two natural people, the C proof above can fail even when $H = 1$ and all radii remain individually bounded.*

PROOF. Let one claimed group contain records of people u and v , use radius $c_{\text{ph}} = \lambda C$, and let u also attain the permitted residual budget $(1 - \lambda)C$. In X , choose the claimed unit's clipped output as $+\lambda Cw$ for a unit vector w . After removing u , v remains in the group. The changed member commitment creates a new canonical label whose clipped output can be $-\lambda Cw$. The claimed path alone then costs $2\lambda C$, not the single λC charge used in Lemma S7. Align the residual change with w to obtain total change

$$2\lambda C + (1 - \lambda)C = (1 + \lambda)C > C \quad \text{for every } \lambda > 0. \quad (9)$$

If an implementation instead keeps the same label despite changed members, then equal labels no longer imply equal outputs and Lemma S2 is invalid. Either representation breaks a premise of the C certificate. The correct fail-closed action is to reject the impure claim before the epoch and use permanent fallback units, not to certify the observed mixed group as a Partial-Handle. $\square$

1.6 Proof of Evidence-Budget Feasibility

This subsection proves the evidence-budget feasibility theorem stated in the main paper.

Call a domain capacity-attaining when separate admissible neighboring pairs attain (i) one complete unit, (ii) H partial and B pure local units, and (iii) H partial and B mixed local transitions, while all other canonical labels cancel.

LEMMA S9 (SUFFICIENCY OF THE THREE INEQUALITIES). *Fix $C > 0$, integers $H, B \geq 1$, and nonnegative uniform route radii. On every H, B -admissible domain, if*

$$c_{\text{ex}} \leq C, \quad Hc_{\text{ph}} + Bc_{\text{atom}} \leq C, \quad Hc_{\text{ph}} + 2Bc_{\text{mixed}} \leq C, \quad (10)$$

then every admissible neighboring pair has WUR at most C .

PROOF. The complete branch follows from the first inequality. For a noncomplete person, Lemma S7 applies. Since $a + m \leq B$,

$$\begin{aligned} ac_{\text{atom}} + 2mc_{\text{mixed}} &\leq (a + m) \max\{c_{\text{atom}}, 2c_{\text{mixed}}\} \\ &\leq B \max\{c_{\text{atom}}, 2c_{\text{mixed}}\}. \end{aligned} \quad (11)$$

The last two inequalities in Eq. (10) are equivalent to $Hc_{\text{ph}} + B \max\{c_{\text{atom}}, 2c_{\text{mixed}}\} \leq C$. Together with $h \leq H$, this proves WUR at most C . $\square$

LEMMA S10 (NECESSITY UNDER CAPACITY ATTAINMENT). *Fix $C > 0$, integers $H, B \geq 1$, and nonnegative uniform route radii. On a capacity-attaining H, B -admissible domain and Q_{sat} , violation of any one of $c_{\text{ex}} \leq C$, $Hc_{\text{ph}} + Bc_{\text{atom}} \leq C$, or $Hc_{\text{ph}} + 2Bc_{\text{mixed}} \leq C$ yields a query with sensitivity greater than C .*

PROOF. For the complete witness, saturate the single old-only complete unit to obtain change c_{ex} . For the pure witness, align the H old-only partial units and B old-only pure units. The Least Sound Envelope theorem gives change $Hc_{\text{ph}} + Bc_{\text{atom}}$. For the mixed witness, each of the B labels has an old and a new commitment. Set every old output to $c_{\text{mixed}}w$, every new output to $-c_{\text{mixed}}w$, and align the partial outputs with w . The attained change is $Hc_{\text{ph}} + 2Bc_{\text{mixed}}$. A violated inequality therefore produces sensitivity above C . $\square$

Completion of the proof. Lemma S9 proves sufficiency and Lemma S10 proves necessity. The maximum form is the conjunction of the pure and mixed endpoint inequalities. Without capacity attainment, the sufficiency proof remains valid, but an absent endpoint witness can make the rectangular conditions stronger than necessary for that restricted domain.

2 Exact Fixed-Allocation Statistics

The evaluation section of the main paper reports rounded fixed-allocation results. The full-precision record uses $\lambda = 0.9$, a one-percentage-point non-inferiority margin, 20 paired configurations per dataset, four methods, four one-sided tests, and Holm correction. UnitScope-Fixed gains 0.02430 [0.01664, 0.03236] over Stable-Local on FEMNIST and 0.04156 [0.03085, 0.05347] on Synthea. Both non-inferiority tests and both gain tests reject after Holm correction with adjusted $p = 3.82 \times 10^{-6}$.

The exact sign-flip tests and the 20,000-draw paired-seed intervals use the frozen analysis. Secondary writer and patient intervals use 5,000 weighted two-level bootstrap draws.

3 Complete Four-Dataset Results

Table S1 supplies the complete account-level results behind the summary in the main paper’s evaluation section. The protocol fixes $C = 1$, $H = 1$, $B = 6$, $\lambda = 0.9$, 50 rounds, a public normalizer of 512, and paired schedules and Gaussian draws. The Sent140 and balanced MovieLens families use $\varepsilon = 4$, $\delta = 10^{-5}$, and noise multiplier 9.215. Sent140 uses 2,000 accounts and a public cap of 32 tweets. The balanced MovieLens task uses 3,000 accounts, a cap of 24 ratings, train-derived deterministic majority thinning, and a 64-hidden-unit multilayer perceptron (MLP). Rating stars create the binary target and never enter the features. Account identifiers are not described as verified natural-person identities.

The Sent140 release contains 1,685 repeated tweet IDs whose duplicate rows carry contradictory labels. The adapter drops all 3,370 ambiguous rows before account selection rather than choosing either label. A stable public hash then selects accounts and caps their records. MovieLens raw data are not redistributed.

Table S1. **The balanced MovieLens task avoids majority-class saturation.** Values are means with 95% paired-seed bootstrap intervals over ten configurations. Complete-ID is an assumption reference.

Dataset	Method	Sensitivity	Records used	Account macro accuracy
Sent140	Stable-Local	C	100.0%	0.5041 [0.4901, 0.5177]
	UnitScope-Fixed	C	100.0%	0.5057 [0.4834, 0.5287]
	Handle-Only	C	26.0%	0.5081 [0.4873, 0.5289]
	Naive-2C	$2C$	100.0%	0.5135 [0.4980, 0.5291]
	Complete-ID Oracle	C	100.0%	0.5337 [0.5030, 0.5623]
MovieLens	Stable-Local	C	100.0%	0.5083 [0.5020, 0.5140]
	UnitScope-Fixed	C	100.0%	0.5107 [0.5054, 0.5152]
	Handle-Only	C	24.9%	0.5111 [0.5055, 0.5157]
	Naive-2C	$2C$	100.0%	0.5108 [0.5042, 0.5167]
	Complete-ID Oracle	C	100.0%	0.5390 [0.5363, 0.5414]

The Sent140 paired interval for UnitScope minus Stable-Local is $[-0.0282, 0.0336]$. The intervals against Handle-Only and Naive-2C also contain zero. On the balanced MovieLens task, both class recalls exceed 0.24 and prediction entropy exceeds 0.51 nats. Exact convergence values appear in Tables S7 and S8.

4 Additional Accuracy and Non-Independent Data Results

This section records additional regimes summarized in the main paper’s evaluation section. The two Synthea privacy-budget families use the same source fingerprint and task window but different splits, topology, evidence, schedule, initialization, and noise seeds. The $\epsilon = 8$ family uses noise multiplier 4.9015, while the $\epsilon = 4$ replication uses 9.2149. A matched-seed comparison on the ten $\epsilon = 8$ Synthea configurations gives paired means of 0.2291 versus 0.2292 for UnitScope and 0.2321 versus 0.2329 for Handle-Only. The corresponding means are 0.2343 versus 0.2359 for Naive-2C and 0.2050 versus 0.2003 for Stable-Local at $\epsilon = 4$ versus $\epsilon = 8$. The signs vary across replicates, so the cross-table reversal reflects family differences and a small stochastic budget effect rather than a monotone privacy-budget law.

Table S2 records the complete comparison.

Table S2. **The wider privacy budget does not uniformly improve utility.** Values are means with 95% paired-seed bootstrap intervals over ten configurations.

Dataset	Method	Records used	Primary metric
FEMNIST	Stable-Local	100.0%	0.1600 [0.1359, 0.1846]
	UnitScope-Fixed	100.0%	0.2159 [0.1689, 0.2644]
	Handle-Only	24.8%	0.2219 [0.1724, 0.2728]
	Naive-2C	100.0%	0.2229 [0.1768, 0.2701]
Synthea	Stable-Local	100.0%	0.2003 [0.1874, 0.2150]
	UnitScope-Fixed	100.0%	0.2292 [0.2129, 0.2458]
	Handle-Only	40.7%	0.2329 [0.2170, 0.2491]
	Naive-2C	100.0%	0.2359 [0.2175, 0.2544]

Validation-tuned non-private reference. The fixed private schedule is not used to estimate model capacity. We therefore keep the same SmallCNN, writer ranks 1000–1999, and the first five split seeds of the frozen replication, but select the non-private learning rate and round count using validation data only. Table S3 records every declared candidate. The selected setting is fixed before one test evaluation on each split.

Table S3. **The SmallCNN can learn the FEMNIST task under a validation-selected non-private schedule.** Grid cells are mean validation writer-macro accuracy over five splits.

Learning rate	50 rounds	100 rounds	200 rounds
0.05	0.1474	0.3343	0.5469
0.10	0.3723	0.4010	0.7475
0.25	0.4788	0.6022	0.9137
0.50	0.2338	0.4322	0.6131
Selected test mean and 95% interval			0.9153 [0.9052, 0.9234]

For the non-independent and identically distributed (non-IID) family, each overlapping FEMNIST writer receives class-conditional Dirichlet silo probabilities. Records are assigned without duplication. Stable non-overlapping writers remain in one silo. UnitScope obtains 0.1414 [0.1144, 0.1694] at $\alpha = 0.1$ and 0.1415 [0.1146, 0.1692] at $\alpha = 0.5$. The decrease from 0.5 to 0.1 is 0.000125, which passes the frozen 0.05 robustness threshold. UnitScope remains below Handle-Only by 0.0055 and 0.0054. This family supports stability across the tested heterogeneity range without supporting an endpoint-superiority claim. Figure S1 reports both trajectories.

5 Final Validity Audit

The final validity audit covers mechanism isolation, measured contribution capacity, privacy accounting, evidence activation on all four datasets, complete-identity degeneration, and logged plan quantities. All six checks pass. Real-data

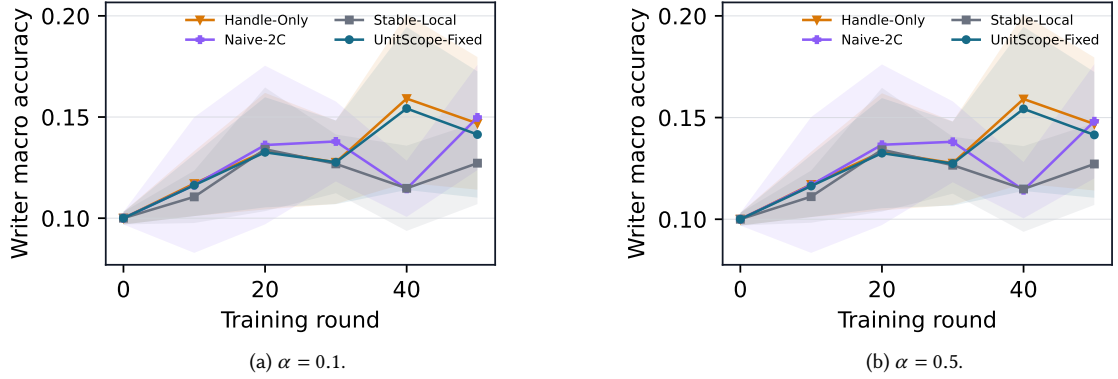


Fig. S1. **User-level non-IID stress test.** Curves report writer-macro accuracy over 50 rounds for five paired configurations per Dirichlet value.

capacities stay below their declared bounds, while a separate legal saturation instance attains C , C , C , and $2C$ for Stable-Local, UnitScope, Handle-Only, and Naive-2C.

6 Frozen Analysis Ledger

The final audit reads four frozen analysis inputs before accepting any result row. Table S4 records their SHA-256 digests. The audit retains 160 successful main runs and 160 successful reference runs, and it reports zero modified frozen runs. These records preserve the declared analysis, exploratory regimes, and failed target checks without rewriting any frozen product.

Table S4. **Frozen analysis and audit digests.** Digests are copied from the final validity record.

Frozen input	SHA-256 digest
Main registry	57827ab5f4cfab6a6fe9159f550011d40735dcde8ff5041d23e416b4e6618df
Reference registry	90bbbd5b3cb3ee9f25bfdfe041c2eab53b713a2a7f84de1bb32ba929f13e2637
Learning protocol	788841b85508b53c9d75b17a29d82de0014859a6ac6f2cfdaa21dba73ee51484
Audit code	2ceb7a46e2b1585d52b5470ab0171acba4df82cbdf78d47783b86608f2e17500

7 Bounded-Component Certificate Example

A structural certificate must cover candidate generation, grouping, and failure routes on every admissible neighbor. A bounded-component argument applies only when vertices are validated pure atomic local units and the candidate hypergraph is public and frozen. Suppose each person removes at most B vertices and maximum degree is $d_{\max}$. If components of size at most K are grouped while larger components remain stable singletons, a conservative label bound is

$$\widehat{\kappa}_{\text{cap}} = 2B(d_{\max} + 1)K + B. \quad (12)$$

Mixed vertices or record-derived edges that change after deletion violate these prerequisites and remain audit-only. Since the bound is at least $3B$, uniform radius $\frac{C}{\widehat{\kappa}_{\text{cap}}}$ is more conservative than pure local fallback $\frac{C}{B}$, so the prototype retains the governed path instead.

8 Reachable-Set and Reserve Derivation

Without capacity attainment, the rectangular feasibility inequalities remain sufficient but may not be necessary. A sufficient fixed-radius condition uses the reachable set $\mathcal{K}_{\mathcal{D}}$.

$$c_{\text{ex}} \leq C, \quad \sup_{(h,a,m) \in \mathcal{K}_{\mathcal{D}}} (hc_{\text{ph}} + ac_{\text{atom}} + 2mc_{\text{mixed}}) \leq C. \quad (13)$$

UnitScope chooses

$$c_{\text{ph}} = \frac{\lambda C}{H}, \quad c_{\text{atom}} = \frac{(1-\lambda)C}{B}, \quad c_{\text{mixed}} = \frac{(1-\lambda)C}{2B}, \quad (14)$$

which makes both noncomplete constraints tight. As λ varies, these radii trace a componentwise Pareto boundary for uniform fixed paths. Adaptive compilers and downstream model accuracy have different frontiers.

For a person with $h < H$ partial groups, fixed paths charge at most

$$\Delta_{\text{fixed}}(h) \leq C - \frac{(H-h)\lambda C}{H}. \quad (15)$$

The term $\frac{(H-h)\lambda C}{H}$ is the minimum *identity-uncertainty reserve* attributable to unfilled handle slots when partial and local maxima are attainable. This capacity remains unused. Reissuing it adaptively requires a separate neighboring-state WUR proof and access-pattern analysis.

9 Operational Atlas

The main paper reports the operational summary table. Figure S2 retains the corresponding compiler, protected-execution, and lifecycle trajectories.

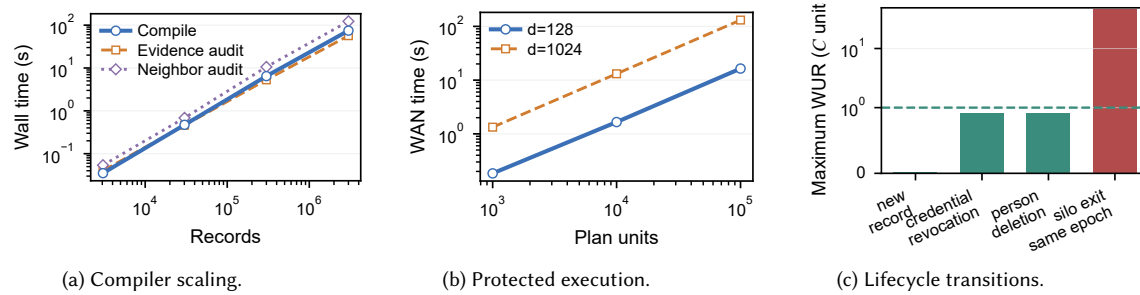


Fig. S2. **Operational atlas.** Compiler time, projected wide-area network (WAN) time, and lifecycle WUR are separated because they measure different execution stages.

10 Exact Data Underlying the Figures

The following tables report the values plotted in the main paper and in the supplementary non-IID figure. Learning-curve tables give the plotted mean at each recorded round. Interval tables retain the same paired bootstrap intervals used by the corresponding figures.

Table S5 records the FEBRL boundary audit.

Table S6 records the NCVR scale audit.

Table S9 supplies the values plotted in Figure S1.

Table S10 supplies the allocation-frontier values.

Table S5. **Exact FEBRL values for the boundary-witness figure.** Progressive Entity Matching (PER) is the external matcher, and WUR is measured in units of C .

Dataset	Linker	Threshold	Pairwise F1	Maximum WUR
FEBRL1	Pilot	0.72	0.9827	5
FEBRL2	Pilot	0.72	0.9375	14
FEBRL3	Pilot	0.76	0.9774	11
FEBRL1	PER	0.72	0.9431	3
FEBRL1	PER	0.76	0.9848	2
FEBRL1	PER	0.80	0.9733	2
FEBRL1	PER	0.84	0.9373	2
FEBRL1	PER	0.88	0.9201	2
FEBRL2	PER	0.72	0.8922	4
FEBRL2	PER	0.76	0.9695	3
FEBRL2	PER	0.80	0.9688	2
FEBRL2	PER	0.84	0.9318	3
FEBRL2	PER	0.88	0.9092	3
FEBRL3	PER	0.72	0.7912	6
FEBRL3	PER	0.76	0.9632	3
FEBRL3	PER	0.80	0.9639	3
FEBRL3	PER	0.84	0.9346	3
FEBRL3	PER	0.88	0.9041	4

Table S6. **Exact North Carolina voter-registration (NCVR) values for the boundary-witness figure.** WUR is measured in units of C .

Scope	People	Threshold	Precision	Pairwise F1	Maximum WUR
County	10,000	0.7	0.9427	0.8640	5
County	10,000	0.8	0.9829	0.8306	4
County	10,000	0.9	0.9968	0.6714	2
County	25,000	0.7	0.8915	0.8333	5
County	25,000	0.8	0.9700	0.8167	4
County	25,000	0.9	0.9960	0.6591	2
County	50,000	0.7	0.8524	0.8098	8
County	50,000	0.8	0.9558	0.8054	5
County	50,000	0.9	0.9941	0.6508	2
Statewide	50,000	0.7	0.9684	0.8757	3
Statewide	50,000	0.8	0.9978	0.8355	3
Statewide	50,000	0.9	0.9998	0.6633	2
Statewide	100,000	0.7	0.8967	0.8433	4
Statewide	100,000	0.8	0.9958	0.8333	3
Statewide	100,000	0.9	0.9996	0.6625	2
Statewide	250,000	0.7	0.7441	0.7571	4
Statewide	250,000	0.8	0.9922	0.8198	3
Statewide	250,000	0.9	0.9993	0.6570	2

Table S7. **Exact learning-curve means for FEMNIST and Synthea.**

FEMNIST writer-macro accuracy				
Round	Stable	UnitScope	Handle	Naive-2C
0	0.1027	0.1027	0.1027	0.1027
10	0.1073	0.1052	0.1058	0.1075
20	0.1155	0.1217	0.1228	0.1222
30	0.1126	0.1230	0.1259	0.1230
40	0.1168	0.1321	0.1349	0.1284
50	0.1306	0.1549	0.1582	0.1398
Synthea patient area under the precision-recall curve (AUPRC)				
Round	Stable	UnitScope	Handle	Naive-2C
0	0.2213	0.2213	0.2213	0.2213
10	0.2068	0.2096	0.2112	0.2110
20	0.2034	0.2178	0.2206	0.2222
30	0.2071	0.2308	0.2351	0.2417
40	0.2122	0.2460	0.2507	0.2573
50	0.2159	0.2575	0.2615	0.2654

Table S8. **Exact learning-curve means for Sent140 and the balanced MovieLens task.**

Sent140 account-macro accuracy					
Round	Stable	UnitScope	Handle	Naive-2C	Oracle
0	0.5023	0.5023	0.5023	0.5023	0.5023
10	0.5001	0.4982	0.5002	0.4976	0.5097
20	0.5001	0.5030	0.5059	0.4970	0.5214
30	0.5043	0.5048	0.5061	0.5029	0.5177
40	0.4991	0.5030	0.5027	0.5095	0.5231
50	0.5041	0.5057	0.5081	0.5135	0.5337

MovieLens account-macro accuracy					
Round	Stable	UnitScope	Handle	Naive-2C	Oracle
0	0.5020	0.5020	0.5020	0.5020	0.5020
10	0.5051	0.5051	0.5053	0.5075	0.5133
20	0.5000	0.4992	0.4989	0.5015	0.5224
30	0.5069	0.5116	0.5117	0.5092	0.5333
40	0.5085	0.5114	0.5122	0.5097	0.5340
50	0.5083	0.5107	0.5111	0.5108	0.5390

Table S9. **Exact non-IID FEMNIST learning-curve means.**

Dirichlet $\alpha = 0.1$				
Round	Stable	UnitScope	Handle	Naive-2C
0	0.1000	0.1000	0.1000	0.1000
10	0.1105	0.1163	0.1171	0.1164
20	0.1341	0.1326	0.1331	0.1362
30	0.1268	0.1276	0.1276	0.1379
40	0.1148	0.1542	0.1591	0.1145
50	0.1272	0.1414	0.1469	0.1498

Dirichlet $\alpha = 0.5$				
Round	Stable	UnitScope	Handle	Naive-2C
0	0.1000	0.1000	0.1000	0.1000
10	0.1111	0.1163	0.1171	0.1168
20	0.1342	0.1325	0.1331	0.1366
30	0.1266	0.1274	0.1276	0.1381
40	0.1149	0.1543	0.1591	0.1143
50	0.1272	0.1415	0.1468	0.1482

Table S10. **Exact values for the allocation frontier.** Means and intervals use ten configurations. The plotted $\lambda = 1$ record-use annotations use the frozen main-study endpoints, 25.0% and 40.3%.

Dataset	λ	Mean	95% interval	Records used
FEMNIST	0	0.1369	[0.1109, 0.1615]	100.0%
FEMNIST	0.1	0.1284	[0.1057, 0.1497]	100.0%
FEMNIST	0.25	0.1354	[0.1099, 0.1595]	100.0%
FEMNIST	0.5	0.1456	[0.1179, 0.1723]	100.0%
FEMNIST	0.75	0.1538	[0.1257, 0.1810]	100.0%
FEMNIST	0.9	0.1588	[0.1314, 0.1856]	100.0%
FEMNIST	1	0.1627	[0.1351, 0.1892]	25.0%
Synthea	0	0.2037	[0.1935, 0.2144]	100.0%
Synthea	0.1	0.1994	[0.1909, 0.2081]	100.0%
Synthea	0.25	0.2072	[0.1955, 0.2194]	100.0%
Synthea	0.5	0.2190	[0.2044, 0.2345]	100.0%
Synthea	0.75	0.2352	[0.2184, 0.2515]	100.0%
Synthea	0.9	0.2454	[0.2265, 0.2625]	100.0%
Synthea	1	0.2506	[0.2317, 0.2678]	40.3%

Table S11 supplies the evidence-coverage values.

Table S12 supplies the gradient-geometry values.

Table S13 supplies the contract-parameter values.

Table S11. **Exact values for the evidence-coverage heatmaps.** Differences and intervals are in percentage points.

q_{user}	$q_{\text{record handle}}$	Δ vs. Stable		Δ vs. Handle	
		Mean	95% interval	Mean	95% interval
0.25	0.25	-4.69	[-6.77, -2.87]	+0.14	[-0.05, +0.34]
0.50	0.25	-1.54	[-2.44, -0.64]	+0.13	[-0.12, +0.35]
0.75	0.25	+0.70	[-0.47, +1.80]	-0.11	[-0.26, +0.08]
0.25	0.50	-1.66	[-2.89, -0.12]	+0.23	[-0.01, +0.44]
0.50	0.50	+1.80	[+0.28, +3.13]	-0.34	[-0.53, -0.10]
0.75	0.50	+6.19	[+2.58, +9.47]	-1.20	[-1.88, -0.48]
0.25	0.75	+0.69	[-0.42, +1.79]	-0.06	[-0.31, +0.19]
0.50	0.75	+5.30	[+2.19, +8.83]	-0.58	[-0.96, -0.21]
0.75	0.75	+10.65	[+5.82, +15.59]	-1.25	[-1.83, -0.66]

Table S12. **Exact noisy-MSE reductions for the gradient-geometry figure.** Values are percentages relative to Stable-Local fallback.

Cosine	$\lambda = 0$	0.25	0.5	0.75	0.9	1
-0.9	+0.0	+16.4	+20.9	+13.3	+5.1	-0.9
-0.5	+0.0	+12.8	+21.3	+17.8	+12.0	+7.7
0.0	+0.0	+11.5	+19.3	+17.5	+12.5	+8.1
0.5	+0.0	+10.3	+17.8	+15.4	+10.5	+6.4
0.9	+0.0	+9.5	+16.5	+13.7	+8.4	+4.2

Table S13. **Exact values for the contract-parameter sweep.** Entries are means with 95% seed-bootstrap intervals.

Handle multiplicity, deterministic MSE				
H	Stable	UnitScope	Handle	
1	5.898e-4 [5.377e-4, 6.477e-4]	2.588e-4 [2.406e-4, 2.782e-4]	2.612e-4	[2.436e-4, 2.777e-4]
2	5.898e-4 [5.380e-4, 6.468e-4]	2.623e-4 [2.429e-4, 2.831e-4]	2.597e-4	[2.412e-4, 2.776e-4]
4	5.898e-4 [5.383e-4, 6.472e-4]	3.050e-4 [2.823e-4, 3.304e-4]	2.868e-4	[2.666e-4, 3.071e-4]
Fallback multiplicity, UnitScope deterministic MSE				
B	Pure local units		Mixed local units	
2	2.558e-4	[2.357e-4, 2.746e-4]	3.369e-4	[3.136e-4, 3.595e-4]
4	3.369e-4	[3.136e-4, 3.595e-4]	3.837e-4	[3.585e-4, 4.092e-4]
6	3.676e-4	[3.429e-4, 3.919e-4]	4.003e-4	[3.740e-4, 4.259e-4]
8	3.837e-4	[3.585e-4, 4.090e-4]	4.087e-4	[3.823e-4, 4.352e-4]
One-release privacy level, noisy MSE				
ϵ	Stable	UnitScope	Handle	
1	2.324e-3 [2.045e-3, 2.639e-3]	2.277e-3 [2.039e-3, 2.519e-3]	2.310e-3	[2.083e-3, 2.529e-3]
2	9.185e-4 [8.008e-4, 1.063e-3]	8.599e-4 [7.664e-4, 9.536e-4]	8.899e-4	[7.979e-4, 9.747e-4]
4	5.530e-4 [4.866e-4, 6.338e-4]	4.886e-4 [4.421e-4, 5.330e-4]	5.168e-4	[4.693e-4, 5.653e-4]
8	4.529e-4 [4.041e-4, 5.084e-4]	3.856e-4 [3.576e-4, 4.142e-4]	4.128e-4	[3.796e-4, 4.507e-4]
16	4.236e-4 [3.823e-4, 4.698e-4]	3.549e-4 [3.326e-4, 3.790e-4]	3.816e-4	[3.505e-4, 4.155e-4]

Table S14 supplies the compiler and execution measurements.

Table S15 supplies the lifecycle and failure outcomes.

Table S14. **Exact values for compiler scaling and protected execution.** WAN time is a projection rather than a network measurement.

Compiler scaling					
Users	Records	Evidence (s)	Compile (s)	Neighbor audit (s)	Peak MiB
1,000	3,058	0.04	0.04	0.05	2.85
10,000	30,160	0.46	0.47	0.68	28.21
100,000	299,958	5.29	6.46	10.59	269.79
1,000,000	3,001,544	56.62	74.43	122.71	2,887.14

Protected execution				
Plan units	Dimension	Local time (s)	Communication (MiB)	WAN time (s)
1,000	128	0.0012	1.96	0.18
1,000	1,024	0.0088	15.64	1.33
10,000	128	0.0077	19.53	1.66
10,000	1,024	0.0940	156.27	13.13
100,000	128	0.0693	195.31	16.40
100,000	1,024	0.9494	1,562.52	131.09

Table S15. **Exact lifecycle and failure-injection outcomes.** A dash denotes a rejected transition or a transition that starts a new accounting epoch before WUR is evaluated.

Lifecycle scenario	Decision	Maximum WUR
Credential reissue without refill	Reject	–
Credential revocation	Fallback without refill	0.9167C
Institution exit, new epoch	New epoch valid	–
Institution exit, same epoch	Requires new epoch	46.4667C
Preallocated new record	Preallocated slot	0.0167C
Single-user deletion	Same epoch safe	0.9167C
Unregistered new record	Reject	–

Injected fault	Prescribed branch followed	Valid plan retained
Cross-domain replay	100%	0%
Expired epoch	100%	0%
Handle quota exceeded	100%	0%
Institution nonresponse	100%	100%
Overlapping claims	100%	0%
Record-count mismatch	100%	0%
Single invalid handle	100%	100%
Unregistered claim	100%	0%
Valid control	100%	100%

Table S16. **Exact values for the $\lambda = 0.98$ endpoint comparison.** Endpoint means come from ten fresh configurations per dataset at $\lambda = 0.98$ and the separately frozen 20-configuration replication at $\lambda = 0.9$. The two rows for each dataset therefore use independent seed and data-partition pools.

Dataset	λ	Configs	Stable-Local	Handle-Only	UnitScope
FEMNIST	0.98	10	0.128451	0.147464	0.147402
FEMNIST	0.9	20	0.130591	0.158151	0.154896
Synthea	0.98	10	0.196580	0.225881	0.225187
Synthea	0.9	20	0.215894	0.261484	0.257451

Dataset	Handle-gap contraction	Stable-gain retention
FEMNIST	98.08%	77.97%
Synthea	82.79%	68.84%

Let U_λ , H_λ , and S_λ denote the endpoint means for UnitScope, Handle-Only, and Stable-Local within the experiment at λ . The reported quantities are $1 - |U_{0.98} - H_{0.98}| / |U_{0.9} - H_{0.9}|$ and $(U_{0.98} - S_{0.98}) / (U_{0.9} - S_{0.9})$, respectively. The experiments use the same dataset fingerprints, task models, 50-round schedule, optimizer settings, privacy parameters, and evaluation metrics, but fresh seeds and data partitions. Stable-Local uses only frozen local units and retains all records. Handle-Only allocates to certified groups and drops residual records.